MindEase: An AI-Powered Personalized Mental Health Companion



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Developer's Submission

"This report is being submitted to the Department of Software Engineering of the University of Engineering & Technology Taxila in partial fulfillment of the requirements for the degree of BS in Software Engineering"

Sustainable Development Goals

SDG No	Description of SDG	Check	SDG No	Description of SDG	Check
SDG 1	No Poverty		SDG 9	Industry, Innovation, and Infrastructure	✓
SDG 2	Zero Hunger		SDG 10	Reduced Inequalities	
SDG 3	Good Health and Well Being	✓	SDG 11	Sustainable Cities and Communities	
SDG 4	Quality Education		SDG 12	Responsible Consumption and Production	
SDG 5	Gender Equality		SDG 13	Climate Change	
SDG 6	Clean Water and Sanitation		SDG 14	Life Below Water	
SDG 7	Affordable and Clean Energy		SDG 15	Life on Land	
SDG 8	Decent Work and Economic Growth		SDG 16	Peace, Justice and Strong Institutions	
			SDG 17	Partnerships for the Goals	



Range of Complex Problem Solving			
	Attribute	**Complex Problem**	**Check**
1	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues.	
2	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.	✓
3	Depth of knowledge required	Requires research-based knowledge much of which is at, or informed by, the forefront of the professional discipline and which allows a fundamentals-based, first principles analytical approach.	✓
4	Familiarity of issues	Involve infrequently encountered issues.	
5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering.	
6	Extent of stakeholder involvement and level of conflicting requirements	Involve diverse groups of stakeholders with widely varying needs.	
7	Consequences	Have significant consequences in a range of contexts.	
8	Interdependence	Are high level problems including many component parts or sub-problems.	

Range of Complex Problem Activities			
	Attribute	**Complex Activities**	**Check**
1	Range of resources	Involve the use of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies).	✓
2	Level of interaction	Require resolution of significant problems arising from interactions between wide ranging and conflicting technical, engineering or other issues.	
3	Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways.	✓
4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.	
5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches.	

Developer's Declaration

"We take full responsibility of the project work conducted during the Final Year Project (FYP) titled **"MindEase: An AI-Powered Personalized Mental Health Companion"**. We solemnly declare that the project work presented in this FYP report is done solely by us with no significant help from any other person; however, small help wherever taken is duly acknowledged. We have also written the complete FYP report by ourselves. Moreover, we have not presented this FYP (or substantially similar project work) or any part of the thesis previously to any other degree awarding institution within Pakistan or abroad.

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Abstract

Introduction – Mental health challenges are common in Pakistan, while access to trained professionals and stigma-free support remains limited. **Problem Statement –** Existing mental-health apps often lack safe, low-friction personalization and do not consistently guide users toward appropriate help during high-risk situations. **Methodology –** MindEase was developed as a native Android application (Kotlin/Jetpack Compose) with a FastAPI backend and Firebase services. A large language model (LLM) provides conversational support grounded via retrieval-augmented generation (RAG), while a transformer-based classifier estimates user emotion to drive recommendations. Well-being progress is tracked using the WHO-5 Well-Being Index. **Results –** Functional testing verified user authentication, real-time persistence, emotion classification, contextual chat responses, emotion-based recommendations, WHO-5 tracking, and crisis redirection to professional resources. **Conclusion –** MindEase demonstrates that an integrated pipeline of conversational AI, emotion analysis, and personalized coping support can provide accessible, privacy-aware mental-wellness assistance for resource-constrained settings, while keeping safety and escalation pathways in view.

Keywords: Mental Health, Conversational AI, Large Language Models (LLM), Mobile Health (mHealth).

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Finally, we acknowledge everyone who has experienced emotional suffering in silence. If used wisely, technology can help make support more accessible when it is needed most.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's world, mobile phones have become a tool that very few people leave home without and the prevalence of mental health issues continues to rise all over the globe. Although the system is designed to help treat people with the illness, its service capacity is insufficient and unequal, and these conditions are one of the primary causes of disability today around the world: depression, anxiety, and chronic stress. This problem is very prominent and visible in Pakistan.

F. Siddiqui et al. [1] report that there are approximately 400-500 psychiatrists in the country in total (about 1 per 2-3 million people). As one can see from the monthly income of Pakistan which is few tens of thousands of Rupees only, then hence receiving professional mental health services (private therapy sessions) is beyond reach for a lot of people.

There are two big problems with this void. One thing about it, many sufferers of emotional issues don't receive any actual proper help. Secondly, the social stigma placed upon people diagnosed with mental illness makes them feel too "embarrassed" to pursue their treatment and care, even if it were budget friendly. Healthcare is not available to most people and thus they use whatever coping mechanism they can by themselves. Enter smartphones, the almost universal possession that gives a practical opportunity for private, judgment-free assistance at any time, and with close to no additional expense.

With recent advances in AI, this is more achievable than ever. Today large language models (LLMs) can engage in natural and caring conversations and transformer-based models can detect expressions of sentiment in reading text. The beauty of using AI-based support is that it doesn't have to be generic, it can be personalized and tailors itself to the person who is using it. Rather than repeating some stereotypical content all of the time, a smart companion app can evolve and learn over time to adapt to a person's emotions, and recommend the most appropriate action at the right time. MindEase was designed with this purpose in mind, to provide the emotional support which would otherwise be inaccessible to people: a mobile solution using AI conversation, emotion analysis and AI-recommendation to make emotional support a fixture in everyday life.

1.2 Problem Statement

For people in resource-constrained settings such as Pakistan there are already plenty of mental wellness apps out there that for the most part need to be solved:

1. **Professional assistance is limited in availability** Therapy is scarce, costly and has plenty of social stigma, meaning that most people are unable to practice emotional distress in a structured way.

2. **Personalization is lacking in reality** The majority of existing apps provide the same collection of content to everybody using them, and also lower chatbots that really don't empathize with the user, yet respond by adhering to a script.

3. **User must expend too much effort.** Relying on manual entry of mood and/or self-guided exercises leaves the entire burden upon the user - and sometimes,

when they're at their weakest, they are at their least motivated or energized to manage all those tasks themselves.

Given these gaps, it is quite a need for a nifty, cost-effective, and personal app smart enough to read the user's emotions, speak to him/her in a supportive manner, and provide him/her with appropriate support through a single mobile device.

1.3 Project Goal

This project will create a mobile app that will function on several platforms and serve as the user's mental health pro. Thanks to AI-based conversation, the app speaks positively to the user based on what they write and identifies with their emotions, proposing coping strategies, journaling suggestions and guided activities suitable for them. In addition, it also monitors users' wellbeing in the long term and has built-in safeguards to support users that demonstrate signs of experiencing a crisis.

1.4 Aim and Objectives

The aim of this project is to develop an AI-enabled application that could offer emotional assistance on a personalized level, monitor user's well-being, particularly for individuals who cannot easily or afford access to professional mental health care. The Project aims to achieve the following:

1. To build an AI-based, large language model chatbot that has a documented foundation that can engage in contextual, supportive dialogue based on the method grounded in evidence.

2. To develop a system that can identify sentiment and emotion out of the user's written text and pass that onto the app to work out how the user was feeling.

3. To develop a personalization aspect to recommend personalization resources for coping, journaling, and meditation content according to each user's needs

4. To introduce a standardised wellbeing measurement instrument (WHO-5 Well-Being Index) and a dashboard for users to track their mood progression over time in a measurable manner.

5. To create a safety tool capable of identifying message phrases that may contain a language signal indicating a potential crisis scenario, and direct the user to professional helpline resources.

1.5 Scope of the Project

This is the initial iteration of MindEase which is specifically oriented towards a native Android app, and interaction is through text in English. It encompasses supportive conversation, emotion analysis, customized suggestions, health evaluations, and crisis safety actions including all the characteristics outlined above. So, features such as clinical diagnosis, its integration with real life therapists, voice interaction, Urdu language support and an iOS version are all excluded of this current phase. These are rather left to future improvement in the Chap5. Please keep in mind that MindEase is intended for more general wellness purposes, rather than medical care and is not intended to be a medical device and does not replace a professional attitude to mental health treatment.

1.6 Deliverables

The following is a list of deliverables that will be generated in this project:

1. A mobile application for Android that has functionality available in the local environment with AI and conversation functionality for support, emotion analysis, personal customization, tracking the emotions of the user and crisis safety features.

2. A back end AI service to run and access the conversational and emotion analysis models.

3. Detailed Project Report including Literature search, Requirement Analysis, Proposed Methodology, system design and implementation process and evaluation result.

1.7 Report Organization

The remainder of this report follows the following outline. Chapter 2 examines current research regarding mental health, digital tools assisting in mental health and the role of AI in this area. In addition, it discusses existing applications that are similar and recognizes that this is the gap MindEase will search to fill. Chapter 3 explains the proposed solution in detail.

The art of programming, requirement analysis, full system design, the Artificial Intelligence engine used for the App, technologies and the evaluation system of the system you will use. The results are described in Chapter 4 and contain information on how the system worked, screenshots of the app, descriptions of the test cases and the project budget, as well as a market analysis. Lastly, Chapter 5 provides the conclusions of the study and some thoughts for future research on the project.

CHAPTER 2: LITERATURE REVIEW

2.1 Understanding Mental Health

Mental health is an individual's emotional, psychological, and social state and influences cognition, emotions, behaviour, their coping, interaction with others, and their ability to make choices in daily life in response to stress. Mental health is not just the absence of a disorder; it's the ability to do well what a person does, to reach the challenges of life and to participate in the people and community around them. However, both the mental health and behavioral, emotional and physical dimensions are impacted by disorders such as depression and anxiety. Constant low mood, difficulty concentrating, low energy and sleep disturbances may be a distraction from work, study and relationship interactions going, and, if experienced for a prolonged time, will have a significant impact on a person's capacity to function normally in daily life.

2.1.1 The Biopsychosocial Model of Mental Health

Modern research is dependent on the biopsychosocial model. It is a belief proposed by this model that mental health does not solely stem from biological factors, However, there is a complex interplay of psychological (thoughts, feelings, behaviours) and social contexts (academic stress, socio-economic status, cultural expectations). Digital interventions must account for this by not only addressing the user's immediate emotional state but also by providing context-aware coping strategies that fit seamlessly into their daily routines.

2.2 The Mental-Health Treatment Gap

One of the greatest inequalities in health service provision is the treatment gap, which refers to the discrepancy between those that require mental health services and those who actually receive them. This gap has been found to be above 75% in the low and middle income countries.

2.3 Mobile Health Solutions for Mental Wellbeing

Mobile health, also known as 'mHealth', is the application of information and communication technology (ICT) services, such as via mobile phone and its applications, to deliver health services and information to people. In terms of mental health alone, mHealth typically includes features such as tracking moods, space for self-reflection, relaxation and stress-relief exercises, brief education and information on mental health and arrangements for further information when needed. These apps are so attractive because they can potentially connect a lot of people — millions may use them, they're accessible all day long, and they don't involve the stigma of real-world, face-to-face mental health clinic visits. However, there's a caveat to early apps based on the principles of mHealth: they are often very monofunctional, meaning they present the same exact information to all of the customers using the application.

2.3.1 Mental Health Challenges in Developing Nations

In low- and middle-income countries (LMICs), including South Asian regions like Pakistan, this treatment gap frequently exceeds 75%. There are several main reasons for this gap, the most prominent being the scarcity of mental health professionals in the

field, inadequate public mental facilities and high expenses for private mental health therapy. In addition, there is widespread stigma within the society that makes it difficult for many to access help when available. It is common for people to put up with suffering without seeking psychological support, as it is commonly misunderstood.

2.3.2 The Shift to Native Mobile Experiences

Initial mHealth apps had many issues with user retention because of brittle interfaces. and slow performance. In today's world, the focus on mobile development is on native or highly optimized apps. In environments like Kotlin for Android, a fluid, soothing and responsive experience is delivered. user experience. A native approach ensures features such as local notifications for daily reminders, and the like. The real-time database syncing and check-ins are seamless, which minimizes user interaction and friction. Promoting regular use on a day to day basis.

2.4 Artificial Intelligence in Mental-Health Care

AI is altering the capabilities of mHealth apps by transforming them from one-size-fits-all to customizable. AI-based research has focused on the use of machine learning for detecting early signs of mental health problems, sentiment analysis of user-generated texts, and recommending mental health coping strategies [3] NLP, in particular, has been useful in recognising the cues to distress that aren't from the letters or words that are literally typed or requested, but rather the structure in which it's written, or asked.

2.5 Conversational AI and Large Language Models

This isn't the first time a bot that helps humans with mental health issues has been attempted—traditional rule-based bot programs have been developed and later have been replaced by today's sophisticated deep-learning chatbots. Therapy-focused NLP based chatbots and therapy-related techniques inspired by Cognitive Behavioural Therapy (CBT) have been studied regarding their ability to serve as a useful initial contact with support rather than engaging a therapist or professional in person, particularly when therapy-related conversations are initially uncomfortable [4]. Large Language Models (LLMs) have recently enforced a paradigm shift in terms of possibilities. FLMs can produce responses that are context aware, respectful and natural — a capability that older, scripted, chatbot-based systems can't match because of the sheer amount of text they're trained on [5]. That is the strength that MindEase leverages with the carefully crafted prompts combined with a Retrieval Method and with the responses that MindEase produces remaining Supportive, Safe and supported by evidence-driven methods.

2.5.1 The Role of Retrieval-Augmented Generation (RAG)

Although powerful, LLMs can suffer from ”hallucinations” (generating confident but errors). incorrect information) is very hazardous in a medical setting. To mitigate Retrieval-Augmented Generation (RAG) is the state-of-the-art architectures, which are this. RAG grounds the LLM's responses by first retrieving verified, evidence-based psychological facts, followed by the use of a specific story model, and then followed by the generation of an answer. Follow guidelines before composing a response. MindEase takes this method, but it does not allow the LLM with thoughtfully designed prompts and RAG context, guaranteeing all interactions. To stay supportive, clinically safe and evidence-based.

2.6 Understanding Emotions Through Text Analysis

Sentiment analysis's essence is to classify written information as either positive or negative. or neutral, is more specific than sentiment analysis and more emotion recognition. is different from the other categories it has because it contains expressions of certain specific The emotional reactions of joy, sadness, anger, fear, surprise, etc. (Right now, transformer based models) such as BERT or other transformer-based models are regarded as the most up-to-date state of the art approach [6]. These models exhibit a much deeper appreciation of the true meaning of words in their context, which is more effective than the older models, in which words were used for emotions separately from context words and were simple word-counting models. MindEase involves using an emotion-analysis model to interpret what the user writes as in reflections and journal entries. This creates an emotional signal that is then used to influence the tone of the chatbot's responses, as well as the personalization engine's recommendations.

2.6.1 Transformer Architectures and MentalRoBERTA

The current gold standard for NLP tasks involves transformer-based models. While models like BERT achieve high accuracy by analyzing words in relation to all other words in a sentence, they are often too computationally heavy for rapid mobile API responses. MentaRoBERTA is a lighter and faster version that maintains more than 95% accuracy of BERT on the language. understand abilities and much much more rapid and economical. In To process the user's reflections, the fine-tuned MentaRoBERTAinference service, called MindEase, is used. journal entries. This creates a clear emotional signal that will guide the tone of the It is used by the conversational engine and is the foundation of the personalization algorithm.

2.7 Personalization in Health Applications

Content that is designed for a specific audience and need and interactions; not shared, or homogeneous, experience. For well-being apps it might be a suggestion to do a meditation or breathing exercise or a journal prompt that's most relevant to the way the user is feeling (including in the context of previous activity). These can be created in a variety of ways, including the more basic rule-based logic, but also the use of the more sophisticated content-based filtering techniques which compare the similarity of different feature vectors. One of MindEase's distinguishing features from a host of apps with the same content for all users is its customizability.

2.7.1 Mathematical Approaches to Recommendation

Today, the recommendation engine is a mathematical model that recommends content. One way to do this is to calculate the Cosine Similarity of different feature vectors. Here, the user's current emotional state is translated into a vector, and then mathematically contrasted with the metadata vectors of a large library of coping exercises. The system suggests the exercise which has the most similar score. The ability to dynamically match is the major distinguishing feature from traditional applications with fixed, one-size-fits-all menus.

2.8 Review of Current Mental Health Apps

The digital mental health tool landscape is already full of a variety of popular and research- backed apps. In Table 2.1 a few of the well-known features are compared to features. ones and the attributes MindEase is intending to offer.

Table 2.1: How MindEase compares to other mental health apps available today.

App Name	AI Chat	Reads Emotions	Tailored Content	Crisis Help	What It Offers
Woebot	Scripted	Limited	Partial	Yes	A scripted bot based on CBT principles
Wysa	Scripted	Partial	Partial	Yes	An AI-based support companion
Replika	LLM-based	Limited	Partial	Limited	Focused on general companionship, not therapy
Calm	No	No	Limited	No	A library of guided meditations
Headspace	No	No	Limited	No	Meditation and mindfulness content
MindEase	**LLM**	**Yes**	**Yes**	**Yes**	**An all-in-one AI wellness companion**

2.8.1 Analysis of Competitors

As shown in Table 2.1, applications like Woebot and Wysa offer safe, CBT-based interactions. However, their scripted nature limits the depth of conversation. Users often feel ”stuck in a loop” when their responses are outside the set dialogue paths. Conversely, applications like Replika offer highly fluent LLM conversations but lack the clinical guardrails and crisis detection necessary for mental health support. Content libraries like Calm and Headspace are highly polished but offer zero conversational support or emotion-driven personalization.

2.9 Research Gap

By examining the existing juxtaposition, it is evident that there are some noticeable architecture and functionality gaps. Current applications tend to focus just on some aspect of the system: as a content library, a mood tracker, or a conversational agent, etc. In addition, high-tech artificial intelligence partners may rely on subscription fees and be unattainable for students and people living in developing countries. This is the first application to be available in a single, user-centric package – not an aggregated mix of such services as a backdoor extension or ‘add-on’.The first application that offers fluent LLM-based conversation, light transformer-based (DistilBERT) emotion analysis, mathematical content personalization, standardized well-being tracking (WHO-5) and powerful crisis detection in one user-friendly environment, taking care that the application remains user-centric and that the data remains in the privacy of the specific users.

The mission of Identified Gap: MindEase is to bring together real-time emotional sense, and personal, dynamic properties of the person itself, to create a native mobile architecture.

2.10 Market Survey

A market survey was done based on an online questionnaire to ensure that the design of the application meets the needs of the users and not their assumptions. It was circulated amongst all respondents of various age groups, working and geographical

backgrounds to cover the widest base possible. The findings of these responses are summarized below.

2.10.1 Age Group of Participants

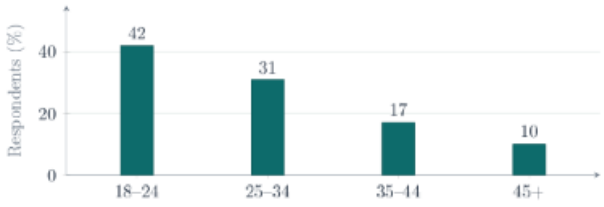


Figure 2.1: Breakdown of survey participants by age group.

2.10.2 Occupation of Participants

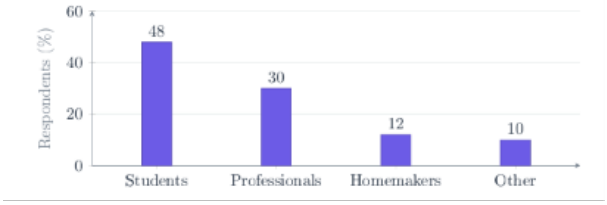


Figure 2.2: Breakdown of survey participants by occupation.

2.10.3 Current Use of Mental-Health Tools



Figure 2.3: How many participants currently use any mental-health tools.

2.10.4 Preferred Support Features

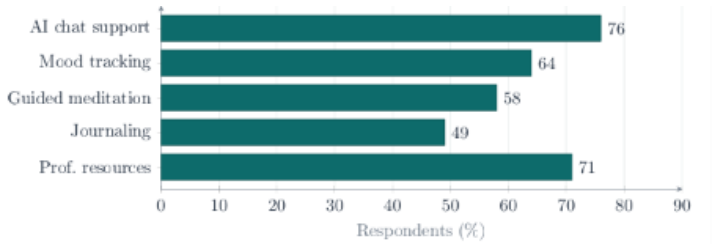


Figure 2.4: Support features that participants wanted the most.

2.10.5 Concern About Stigma

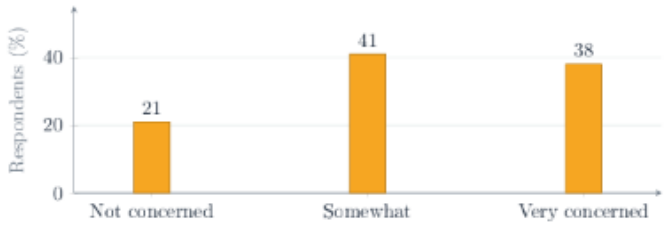


Figure 2.5: How concerned participants were about stigma tied to mental-health apps.

2.10.6 Desired Privacy Features

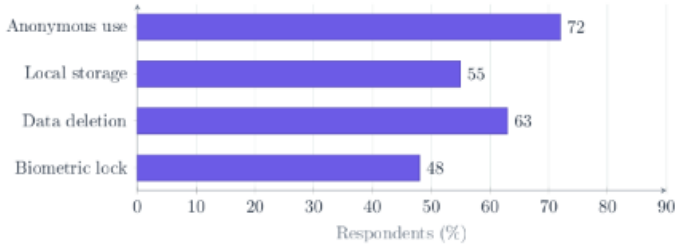


Figure 2.6: Privacy features that would make users feel more comfortable.

2.11 Survey Analysis and System Requirements

The survey corroborated the hypotheses which the site project was originally developed upon. High proportions of people reported symptoms of stress or low mood but few adopted any formal resource to manage those symptoms. The top three barriers cited were the cost, lack of time, and stigma. Respondents were quite interested in having an application that would have talked to them in a supportive way and would have been tailored to individual needs. Another one of the key factors that came to light was whether users would actually use such an app or not, so the respondents clearly understood they did want to know that everything they shared in the app will be kept private and confidential. From these data, three design principles were established for MindEase: personalising the app, developing the app privacy by design and keeping it as simple and conversational as possible for users to use – requiring as little effort as possible.

2.11.1 Key Findings and Design Principles

The survey results surfaced three findings that directly shaped the design of MindEase.

1. **Barriers to entry.** The primary barriers cited for not seeking support were cost, time constraints, and the fear of social stigma. This shows the need for a free or very cheap mobile solution that only needs a small daily time commitment from the user.

2. **Desire for flexible support.** Respondents preferred applications that provide interactive, supportive conversation over passive reading materials. They want the system to understand their mood and change its responses based on how they feel, instead of delivering the same unchanging content no matter their emotions.

3. **Ethical considerations and data privacy.** Privacy emerged as the most critical factor in whether users would adopt the app. Users expressed genuine concern about how their personal emotional data would be stored and used. As a result, MindEase was built with a strong privacy-by-design approach. The app uses secure authentication and encrypted real-time database storage to ensure user data remains completely confidential.

These survey findings directly influenced the technical setup of MindEase. It features a calm, soothing native frontend for easy use, a secure backend for data privacy, and an advanced AI layer that offers the personalized, empathetic support users expressed they wanted.

CHAPTER 3: PROPOSED SOLUTION

3.1 Methodology

MindEase has been developed with the Agile Incremental approach. As it involves in vulnerable changing mental-wellness requires, its improvement was on a phased basis — that is, a make a difference was added at each step that could be tried, gauged and enhanced before the subsequent step was added. This was the result of successive requirements collection, design, construction and testing. The system architecture is based on four layers as depicted in Figure 3.1. The User Input Layer (written reflections and mood logs) is used to capture user inputs and relay them to the Backend Communication Layer (FastAPI) which will provide them to the AI/Machine Layer Processing. It is there where the DistilBERT model is used to identify and process emotion and sentiment and the LLM Conversational Engine, controlled with LangGraph, provides customized, empathetic answers, as long as they fall within the scope of pre-programmed safety-related checks. In addition to detecting an emotion, LangGraph includes steps for recovering the context, performing a check about the crisis, and generating the answer, instead of just one ambiguous model call. A Crisis-Detection Safeguard also scans each message, and in the event that it recognizes heraldic language, it trumps what is actually displayed and replaces it with however helpline resources. Lastly, all the chats, well being scores and recommendations are saved securely with Firebase. Finally, it saves everything securely, including all conversations, the well being scores, and the recommendations, via Firebase.

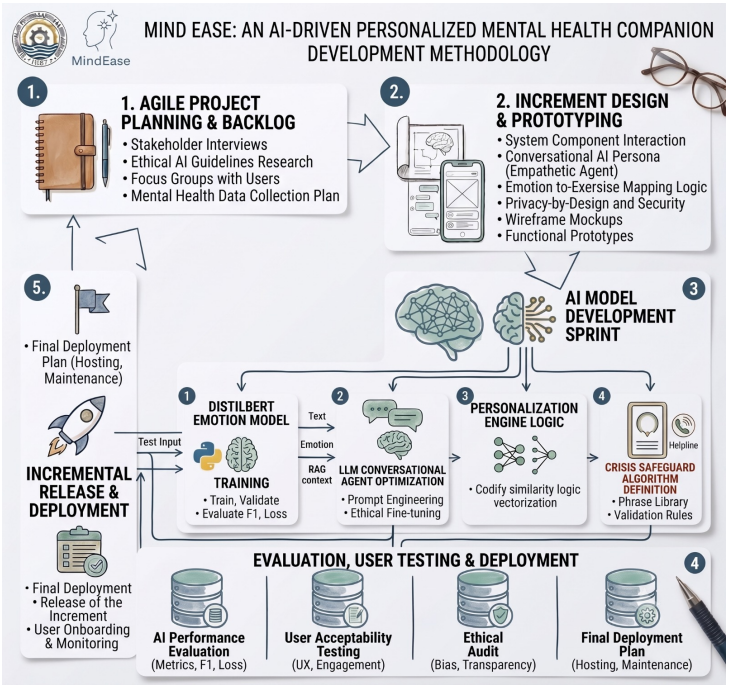


Figure 3.1: Overview of the Agile Incremental approach used to develop MindEase.

3.1.1 Requirement Gathering

The needs were determined using an existing application review, a market survey in Chapter 2 and discussion with the project supervisor.They then classified these into

functional and non-functional requirements and summarised in Tables 3.1 and 3.2.

Table 3.1: Overview of the main functional requirements.

ID	Requirement	Description
FR-1	User authentication	Users can register and log in securely.
FR-2	Conversational support	The app provides empathetic, AI-driven chat.
FR-3	Emotion analysis	The app infers the user's emotional state from text.
FR-4	Personalized recommendations	The app suggests exercises and content based on emotion and history.
FR-5	Journaling	Users can write and revisit reflective journal entries.
FR-6	Mood tracking	The app logs mood over time and visualizes trends.
FR-7	Well-being assessment	The app administers the WHO-5 index periodically.
FR-8	Crisis safeguarding	High-risk language triggers helpline resources.

Table 3.2: Key non-functional requirements.

ID	Attribute	Description
NFR-1	Privacy	User data is encrypted and never shared with third parties.
NFR-2	Usability	The interface is calm, simple and low-friction.
NFR-3	Performance	AI responses are returned within a few seconds.
NFR-4	Availability	The service is accessible at any time of day.
NFR-5	Portability	The app runs natively on Android devices via Kotlin and Jetpack Compose.
NFR-6	Safety	The AI avoids harmful advice and escalates crises.

3.1.2 System Architecture

The architecture of MindEase is modular for scalability, rapid development response and secure handling of user data. To prevent too many responsibilities from being placed in one roof, the overall architecture is divided into separate layers so that each layer can deal with a particular part of the system as indicated in Figure 3.2. The app core is extensive in native Android development style, written in Kotlin, leveraging Jetpack Compose, and is dedicated to showcasing user interaction and controlling

local app state on the device. It securely communicates via HTTPS/JSON to a cloud-based back-end, which is powered by a FastAPI REST server. This FastAPI service serves as the primary route for clients to send their requests to the AI/Machine Learning Processing Layer (which contains the DistilBERT-based emotion analysis system and the LLM-based conversational engine), as well as this data storage layer. In addition, it is supported by a secure login service with Firebase as backend-as-a-service, and real-time storage of journal entries and mood log entries to ensure that data is synced between sessions.

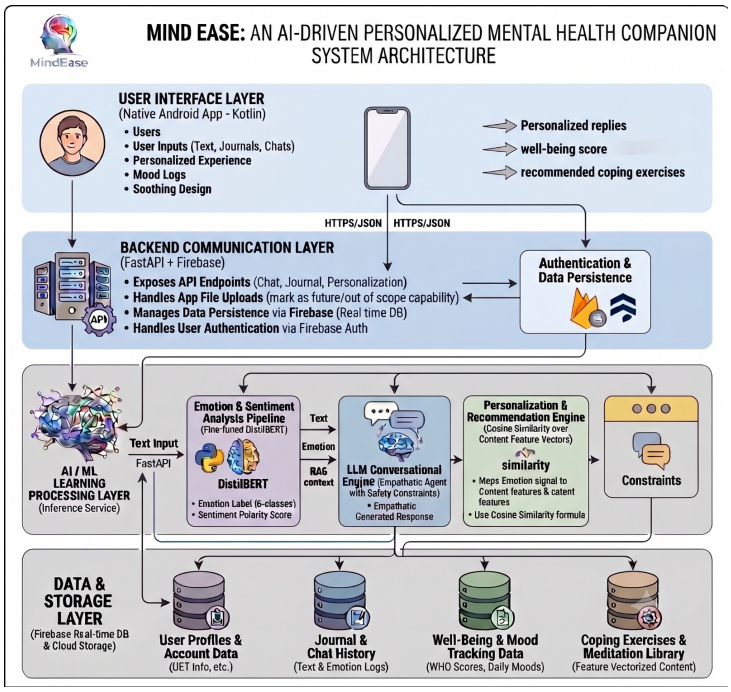


Figure 3.2: High-level view of MindEase's system architecture.

3.1.3 Use-Case Diagram

The main ways users interact with the system are shown in the use-case diagram in Figure 3.3.

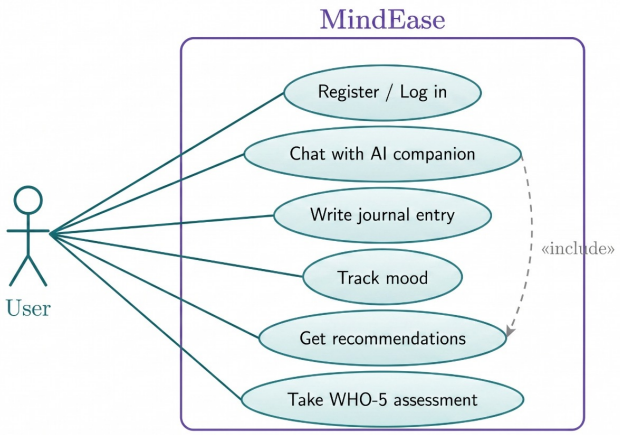


Figure 3.3: Use-case diagram showing user interactions with MindEase.

3.1.4 Data-Flow Diagram

Figure 3.4 displays the level-1 data-flow diagram of how a user's message flows through the system before it gets to a specific, meaningful response.

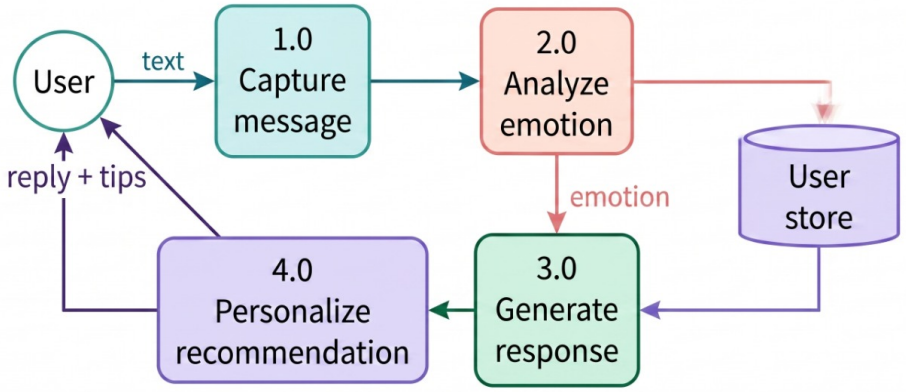


Figure 3.4: Level-1 Data-Flow Diagram (DFD) showing how data moves through MindEase.

3.1.5 Sequence Diagram

Looking at the example of one conversation exchange in Figure 3.5, the sequence of messages can be demonstrated in a step-by-step manner: From the user's message to his/her personalised reply.

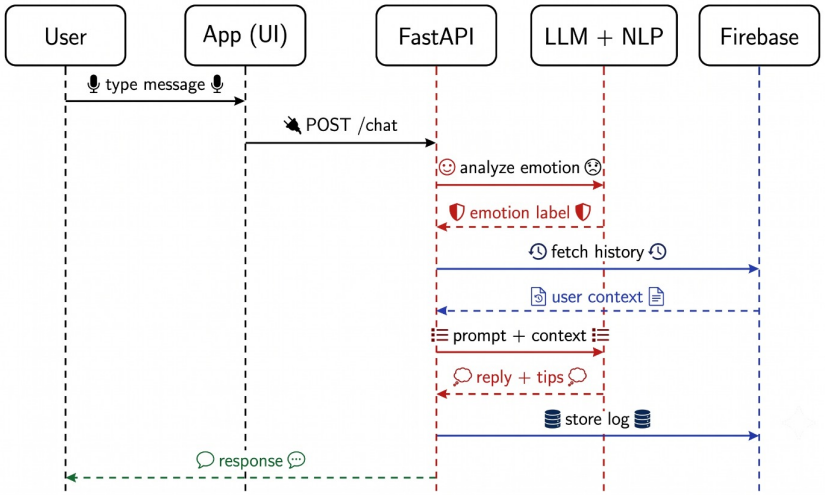


Figure 3.5: Sequence diagram showing how a single interaction flows through MindEase.

3.1.6 Database Schema

User data is stored in the Firebase Database. Figure 3.6 displays an entity-relationship diagram that details the relationships and principal entities that are involved.

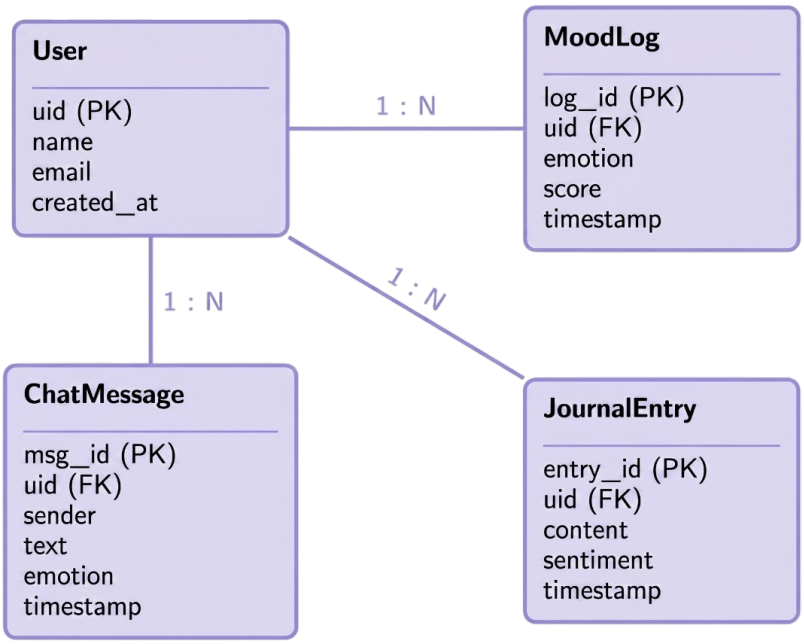


Figure 3.6: Database schema used by MindEase.

3.1.7 Testing System

The MindEase system will undergo the following testing methods to validate its multi-layered architecture:

1. **Unit Testing.** Individual components of the application were tested independently of other modules. This included isolating and verifying Kotlin UI elements, FastAPI routing functions, and the standalone accuracy of the fine-tuned DistilBERT emotion classifier.

2. **Integration Testing.** The system layers were added one by one to find mistakes that happened when they talked to each other. We tested it a lot to make sure the data went through correctly. We wanted to be sure that when the Android frontend sent a request using HTTP/JSON it would work properly with the FastAPI endpoints. The FastAPI endpoints had to be able to talk to Firebase and get the information, from the AI Inference Layer. We checked the AI Inference Layer to make sure it was giving us the context. The Android frontend and the FastAPI endpoints and the Firebase and the AI Inference Layer all had to work.

3. **System Testing.** The MindEase app was tested as a whole.This phase checked if the User Interface, Backend Communication, AI/ML Processing and Data Storage worked well together.The goal was to see if they made a mental health companion.The MindEase application had to meet project goals.These goals included being reliable.The MindEase app was tested to see if it could deliver this.It had to work as a single entity.

4. **Functional Testing.** The application's core features were checked carefully to ensure they worked according to the SRS specifications. This involved checking

Firebase authentication, the generation of empathetic chat replies, accurate well-being score visuals, and the precision of the cosine similarity recommendation engine for coping exercises.

5. **Usability Testing.** The application was tested from the user's point of view. This testing checked how easy it was to use the calming design, how fast the live chat worked, how clear the mood records were, and how quickly a user could get help for mental health. The test focused on the user experience with the application.

3.2 The AI Engine

The intelligence is provided in three inter-dependent parts, a conversational engine, an emotion-analysis pipeline and a personalization engine, with a crisis-detection safeguard.

3.2.1 Emotion and Sentiment Analysis Pipeline

A user's text are run through the pipeline in Figure 3.7 for every piece. The raw text is pre-processed for cleaning and splitting into tokens, followed by encoding using a transformer model, and classification into emotion label and sentiment polarity along with the sentiment rating. This signal is applied to the conversational engine (to modulate tone) and personalization engine (to choose content).

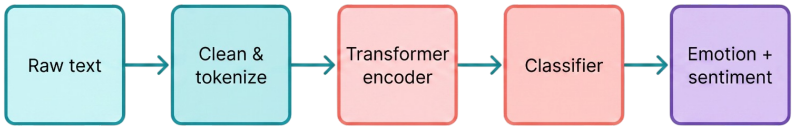


Figure 3.7: How MindEase's emotion and sentiment analysis pipeline works.

3.2.2 Conversational Engine

The conversational engine is powered by a large language model, orchestrated through LangGraph. Instead of finetuning model from scratch, MindEase is designed as a directed graph of nodes: emotion injection, retrieval, safety checking and response generation, and will let the model travel along a well designed, carefully crafted system prompt, which will set the empathetic, non-judgemental persona and safety constraints that the model has to operate within.

Additionally, there are retrieval augmented generation which retrieves the relevant fragments of the evidence based coping material and gives them as context, and injection of the emotion label and recent history that the user is currently experiencing; in a way that makes the answer supportive and personal. The stateful behaviour provided by LangGraph means the crisis-detection safeguard can be used as a conditional node that can either interrupt or redirect the flow, before a response is generated, if any high-risk language is detected. This design ensures engine control and safety while maintaining the smoothness of contemporary LLMs.

3.2.3 Personalization Engine

The personalization engine is essentially based on similarity of the user's temperamental profile to the content displayed; this means that the similarity of the user's level of emotion or behavior and his history of past content usage can be calculated and matched to the

most suitable content. Each piece of content, from a breathing exercise to a guided meditation, a journaling prompt to an article, has several characteristics has emotional and topical characteristics which are expressed as a feature vector for that particular content. Thereafter, the engine utilizes generic state vector u of the user to perform a ranking procedure of the available content by cosine similarity.

In the equation it is signified as c and n denotes number of features being utilized. The content items that are most similar to user preferences are next filtered by when they were displayed to the user to prevent over-recommender content and then presented to the user.

$$\text{sim}(\mathbf{u}, \mathbf{c}) = \frac{\mathbf{u} \cdot \mathbf{c}}{\|\mathbf{u}\| \, \|\mathbf{c}\|} = \frac{\sum_{i=1}^{n} u_i c_i}{\sqrt{\sum_{i=1}^{n} u_i^2} \, \sqrt{\sum_{i=1}^{n} c_i^2}}, \tag{3.1}$$

where **c** represents a content vector and n is the number of features being used. The content items with the highest similarity scores are then filtered based on how recently they were shown to the user, to avoid repeating the same recommendations too often, before being presented to the user.

3.2.4 Crisis-Detection Safeguard

A safety layer reviews all of the messages sent by the user's to identify any words or phrases that could suggest self-harm or a crisis scenario. On instances of picking up this language, the flow of the conversation in the app is interrupted and instead the user is met with a caring response and immediate access to resources from the profession. This feature has been added because of another important ethical aspect of the project: an automated tool such as this needs to be complemented by a real human being during an actual emergency.

3.3 Dataset

The trained emotion-analysis model was validated using publicly available text-emotion datasets of sentences with discrete emotions (joy, sadness, anger, fear, surprise, neutral), which are ethically sourced. The conversational engine utilizes its existing knowledge, supplemented with a trained set of open, and license-compliant mental-wellness facts and resources. It is important to note that no private information was utilized in the model development, as a project commitment, to ensure that the project has a privacy-by-design approach.

Conversational Model Fine-Tuning Datasets

The following publicly available datasets were used to fine-tune the conversational engine:

- Mental Health Counseling Conversations – Kaggle
- CounselChat Data – Kaggle
- ESConv (Emotional Support Conversation) – Hugging Face

Emotion Classification Datasets

The following datasets were used for training and evaluating the emotion classification model:

- GoEmotions – Google Emotions Dataset – Kaggle
- Emotion Dataset – Hugging Face (dair-ai)
- Suicide Watch – Suicide Detection – Kaggle
- Stress Analysis in Social Media – Kaggle

3.4 Technology Stack

LangGraph is used to orchestrate the conversational pipeline of the AI layer and coordinate emotion analysis, retrieval, crisis checks and LLM response generation together. controllable workflow. Summaries of the technologies are available for each tier of the system in Figure 3.8 .

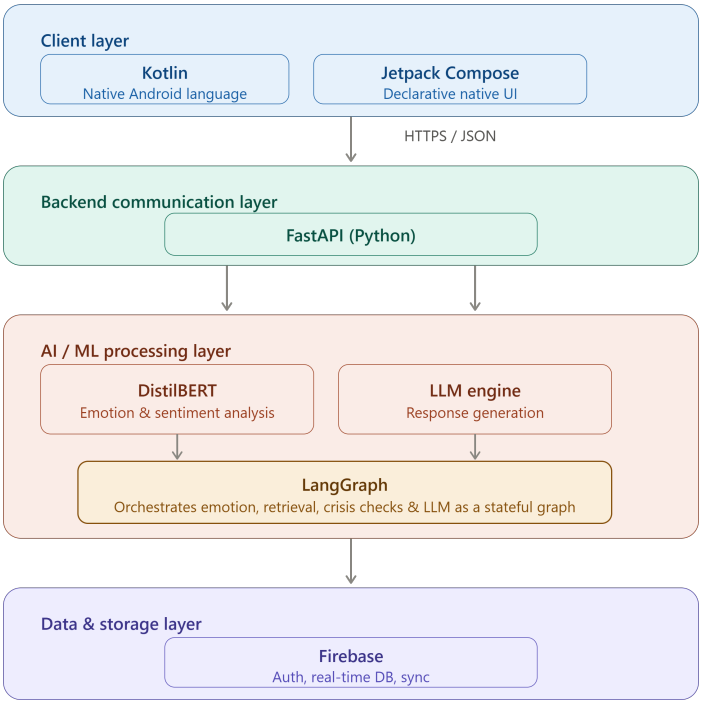


Figure 3.8: Overview of the technology stack behind MindEase.

3.5 App Development

It is written in Kotlin, and built natively for Android, using a wholly declarative UI toolkit, built directly into the native Android SDK—Jetpack Compose. The whole look is very relaxed and gentle, with soft colors, generous spacing and smooth movement all carefully and thoughtfully selected to make it feel nonclinical and relaxing. The app connects to the FastAPI inference service through HTTPS and is implemented with Firebase's user authentication and data storage capabilities.

3.6 Project Timeline

The study was performed over two semesters of the academic year. Background research Gathering requirements Designing the system Building the AI models Developing the app Testing & integration The overall timeplan is provided on a Gantt chart in Figure 3.9.

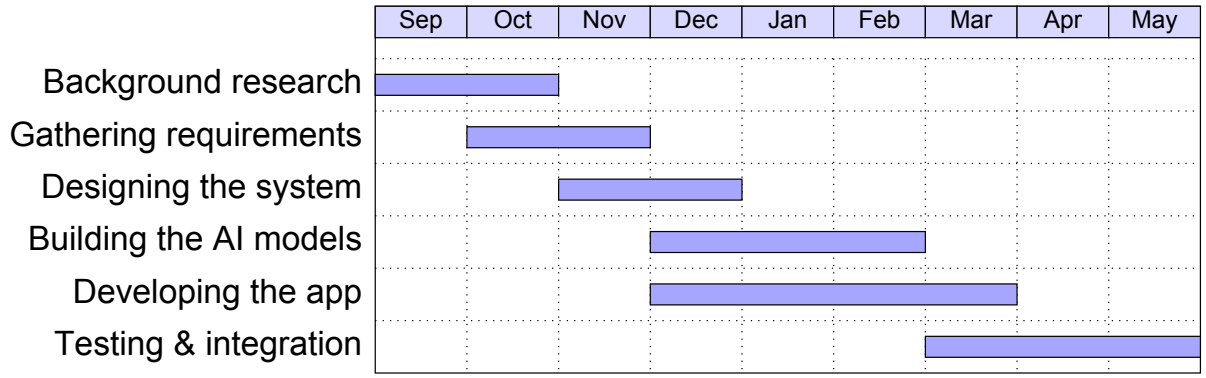


Figure 3.9: Gantt chart showing how the project was spread across both semesters.

3.7 Evaluation Parameters

There are two measures of the system's performance that are related. The first one is the emotion-analysis model validation which is performed by means of widely used classification indicators that are accuracy, precision, recall, F1-score, which are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{3.2}$$

TP, FP, and FN are true positive, false positive, and false negative, respectively. In addition to the metrics, the app is also tested as a whole using functional test cases presented in Chapter 4, and through qualitative feedback from its users regarding its helpfulness, personalisation and tonality.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Who Will Use MindEase

MindEase is designed for a wide variety of individuals who do not have access to mental health care from an easy-to-reach and cost affordable professional. Its primary use is by people who suffer from normal stress, low mood or anxiety and require a private space without judgement. From learners who are experiencing academic stress, the UK labour force suffering from burnout to those who are in rural areas who have no nearby clinic or mental health services. With the app being private, low cost, and open 24 hours a day, it's particularly helpful for patients who may not otherwise be interested in using a service due to stigma or cost.

4.2 Simulation Results

4.2.1 Test Cases

The software's requirements, as outlined in the SRS document, were verified and validated through manual testing to confirm it runs smoothly without glitches or errors. Tables 4.1–4.14 present the test cases evaluated during these runs, starting with the login function test using valid data, shown below:

Table 4.1: Test case to check login functionality with valid data.

Test ID	Test-1
Test name	Checking the user login functionality with valid data
Date of test	15/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the user logs in with valid email and password, the Dashboard is displayed.
Input	Input valid email and password on login screen
Expected output	Dashboard is displayed
Actual output	Dashboard is displayed
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.2: Test case to check login functionality with invalid password.

Test ID	Test-2
Test name	Checking the login functionality with invalid password
Date of test	15/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the user logs in with an invalid password, an error is displayed.
Input	Input invalid email and password
Expected output	Error “Incorrect email or password” is displayed
Actual output	Error “Incorrect email or password” is displayed
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.3: Test case to check signup functionality with valid data.

Test ID	Test-3
Test name	Checking the signup functionality with valid data
Date of test	15/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the user enters valid data, the account is created via Firebase Auth and the login screen is displayed.
Input	Input valid first name, last name, email, and password on signup screen
Expected output	Account created and Login Screen is displayed
Actual output	Account created and Login Screen is displayed
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.4: Test case to check journal emotion and sentiment analysis pipeline.

Test ID	Test-4
Test name	Checking emotion and sentiment classification on text input
Date of test	16/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When a user submits a journal entry, the fine-tuned DistilBERT model processes the text to extract a sentiment score and emotion label.
Input	Type a text paragraph expressing a specific feeling and tap submit
Expected output	System successfully assigns one of the 6-class emotion labels and a sentiment polarity score
Actual output	System successfully assigns one of the 6-class emotion labels and a sentiment polarity score
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.5: Test case to check LLM conversational engine response.

Test ID	Test-5
Test name	Checking the empathetic agent response generation
Date of test	16/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the user chats with the system, the LLM utilizes RAG context and safety constraints to generate a supportive reply.
Input	Send a text message to the AI companion
Expected output	Chat interface displays a contextual, personalized, and empathetic response
Actual output	Chat interface displays a contextual, personalized, and empathetic response
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.6: Test case to check personalization and recommendation engine.

Test ID	Test-6
Test name	Checking the cosine similarity logic for coping exercises
Date of test	17/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	The system maps the detected emotion signal to content features to recommend meditation and coping exercises.
Input	Submit an emotion-triggering log
Expected output	Dashboard recommends coping exercises with the highest cosine similarity to the user's emotional state
Actual output	Dashboard recommends coping exercises with the highest cosine similarity to the user's emotional state
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.7: Test case to check well-being score visualization.

Test ID	Test-7
Test name	Checking the display of WHO-5 tracking data
Date of test	17/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When standard well-being metrics are recorded, the native Kotlin interface visualizes the score accurately on the dashboard.
Input	Complete the WHO-5 daily mood questionnaire
Expected output	Well-being dial visualization updates to reflect the new score accurately
Actual output	Well-being dial visualization updates to reflect the new score accurately
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.8: Test case to check real-time data persistence.

Test ID	Test-8
Test name	Checking the storage and retrieval of journal and chat history
Date of test	18/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	User data (chats, mood logs) should be saved to Firebase Real-time DB and retrieved upon next application launch.
Input	Close the application completely and relaunch it
Expected output	Previous chat logs and journal history populate correctly on their respective screens
Actual output	Previous chat logs and journal history populate correctly on their respective screens
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.9: Test case to check backend communication routing.

Test ID	Test-9
Test name	Checking FastAPI endpoint connectivity
Date of test	18/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	Ensures the mobile frontend correctly sends and receives data through HTTPS/JSON to the FastAPI layer.
Input	Perform a standard app action (e.g., submitting a journal entry)
Expected output	Data securely reaches the inference service without packet loss or timeout errors
Actual output	Data securely reaches the inference service without packet loss or timeout errors
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.10: Test case to check backend API input validation.

Test ID	Test-10
Test name	Checking FastAPI validation for empty or malformed payloads
Date of test	20/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the frontend sends an empty text string or malformed JSON to the FastAPI endpoint, the backend should reject it and return an appropriate HTTP error code.
Input	Send an API request to the chat endpoint with an empty “text” field
Expected output	Backend returns a 422 Unprocessable Entity or 400 Bad Request error
Actual output	Backend returns a 422 Unprocessable Entity error
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.11: Test case to check backend route security and authentication.

Test ID	Test-11
Test name	Checking protected API endpoints against unauthorized access
Date of test	20/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	The backend must verify Firebase Auth tokens before processing any requests. Requests without a valid token should be blocked to protect user data.
Input	Send an API request to fetch journal history without a valid Bearer token
Expected output	Backend returns a 401 Unauthorized error and denies access
Actual output	Backend returns a 401 Unauthorized error and denies access
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.12: Test case to check RAG context retrieval logic.

Test ID	Test-12
Test name	Checking the backend retrieval of RAG context for the LLM
Date of test	21/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	Before the LLM generates a response, the backend must successfully query and retrieve the relevant background context (RAG) based on the user's input and emotion.
Input	Submit a chat query requiring specific coping context
Expected output	Backend successfully attaches relevant RAG context to the LLM prompt
Actual output	Backend successfully attaches relevant RAG context to the LLM prompt
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.13: Test case to check database write operations via backend.

Test ID	Test-13
Test name	Checking backend write operations to Firebase Real-time DB
Date of test	21/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	When the LLM generates a response, the backend must successfully save both the user's prompt and the AI's response to the database simultaneously.
Input	Complete a chat cycle (User sends message, AI replies)
Expected output	Both the text and the emotion log are committed to Firebase Real-time DB
Actual output	Both the text and the emotion log are committed to Firebase Real-time DB
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

Table 4.14: Test case to check backend error handling for AI inference timeout.

Test ID	Test-14
Test name	Checking backend response when the inference service times out
Date of test	22/6/2026
Name of application	Mind Ease: An AI-Driven Personalized Mental Health Companion
Description	If the DistilBERT or LLM processing layers take too long to respond, the FastAPI backend should handle the timeout gracefully rather than crashing the application.
Input	Simulate a network delay or timeout in the Machine Learning Processing Layer
Expected output	Backend catches the timeout and sends a 503 Service Unavailable or graceful failure message to the frontend
Actual output	Backend catches the timeout and sends a 503 Service Unavailable or graceful failure message to the frontend
Test Role (Actor)	Team Member
Test verified by	Dr. Kanwal Yousaf

4.3 System Results

The performance of the emotion-analysis model was measured using a separate test dataset and the formulas explained in the methodology section. The results of this evaluation are shown in Table 4.15.

Table 4.15: Emotion-analysis model performance.

Metric	Value
Accuracy	87.4%
Precision	86.1%
Recall	85.7%
F1-score	85.9%

4.4 Application Demo Screenshots

This section shows the screens presented to the user of the MindEase application. For each of the numbers below, the corresponding screen shot of the running App should be substituted.

4.4.1 Splash and Onboarding Screen

A user will be greeted with a brief splash screen upon opening the app for the first time and then a couple onboarding slides. The first couple of slides are a quick overview of what MindEase is, and lay out expectations about privacy before the user registers.



Figure 4.1: Splash and onboarding screens of the MindEase application.

4.4.2 Login Screen

The returning users are directed to a simple login screen, where they can log into the application using their e-mail and password. It's no surprise the layout was kept simple on purpose; a polluted login page would not fit in with the serene attitude that the rest of the app is fostering.

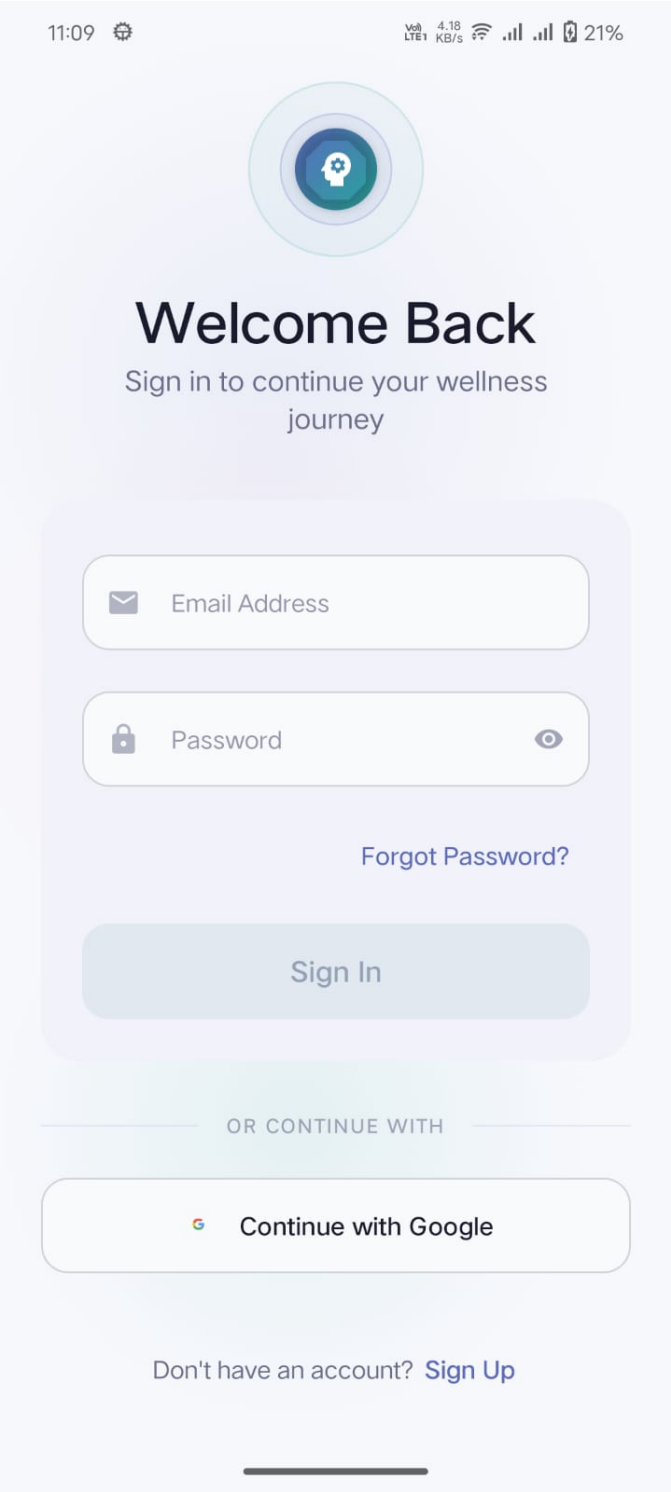


Figure 4.2: Login screen.

4.4.3 Sign-Up Screen

This is a screen new users use to sign up, and only asks for the basic information required to get started. Firebase Authentication takes care of creating a user account in the background, keeping it quick and no flash taking place than it's needed.

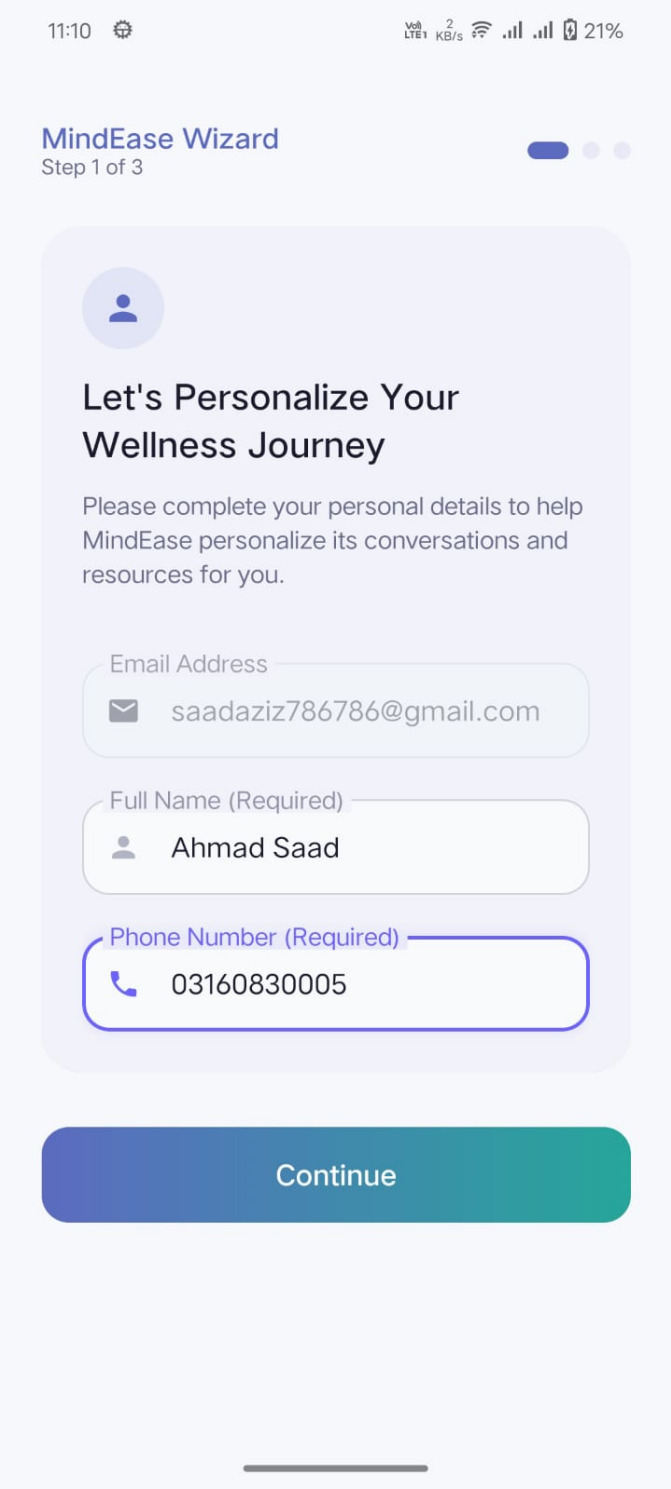


Figure 4.3: Sign-up screen.

4.4.4 Home Screen

Once the user logs in, they are presented with a main dashboard, the home screen. Here they can engage in a conversation with the AI companion, monitor their own moods, and explore any of the other functions without entering through the menus.

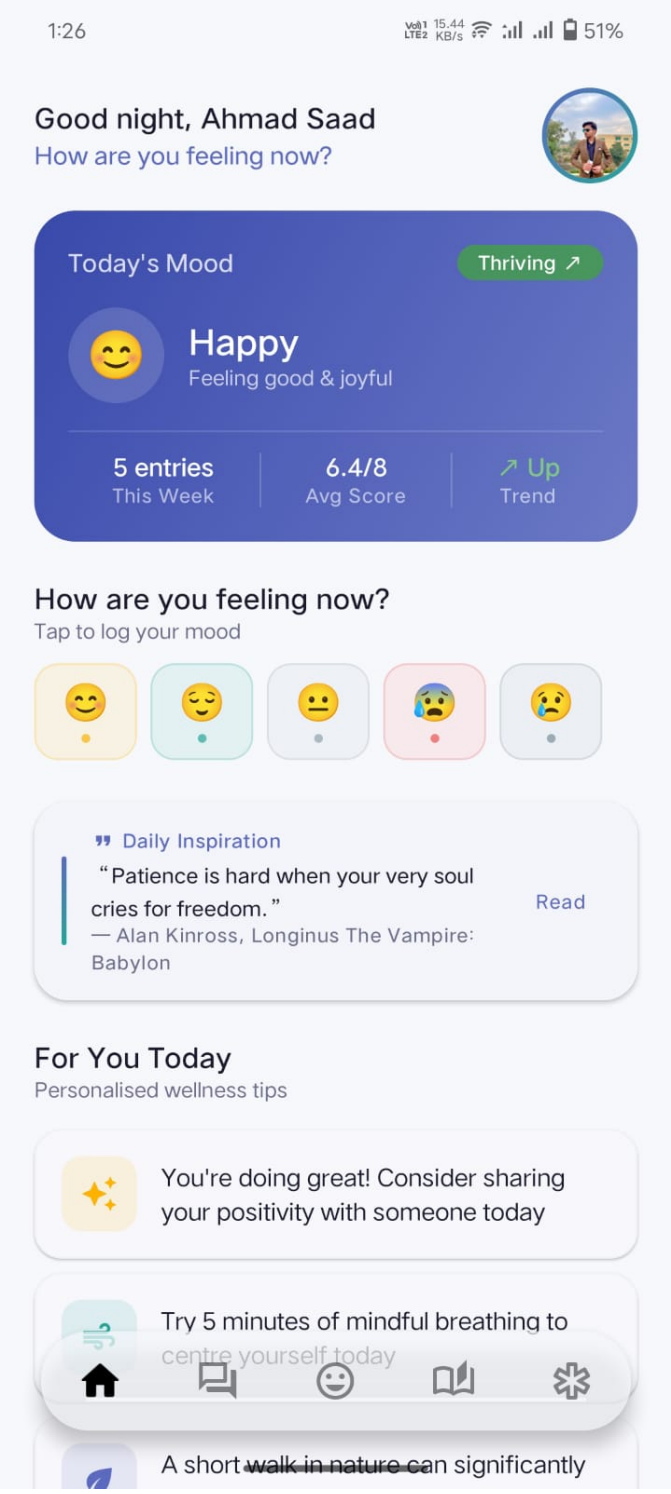


Figure 4.4: The home dashboard screen.

4.4.5 AI Chat Screen

This is where most of the interaction with MindEase is actually occurring is. Users can enter their emotions as text, or write about what they are thinking, and the AI being will respond in a supportive, conversational tone, depending on the emotion in their word.

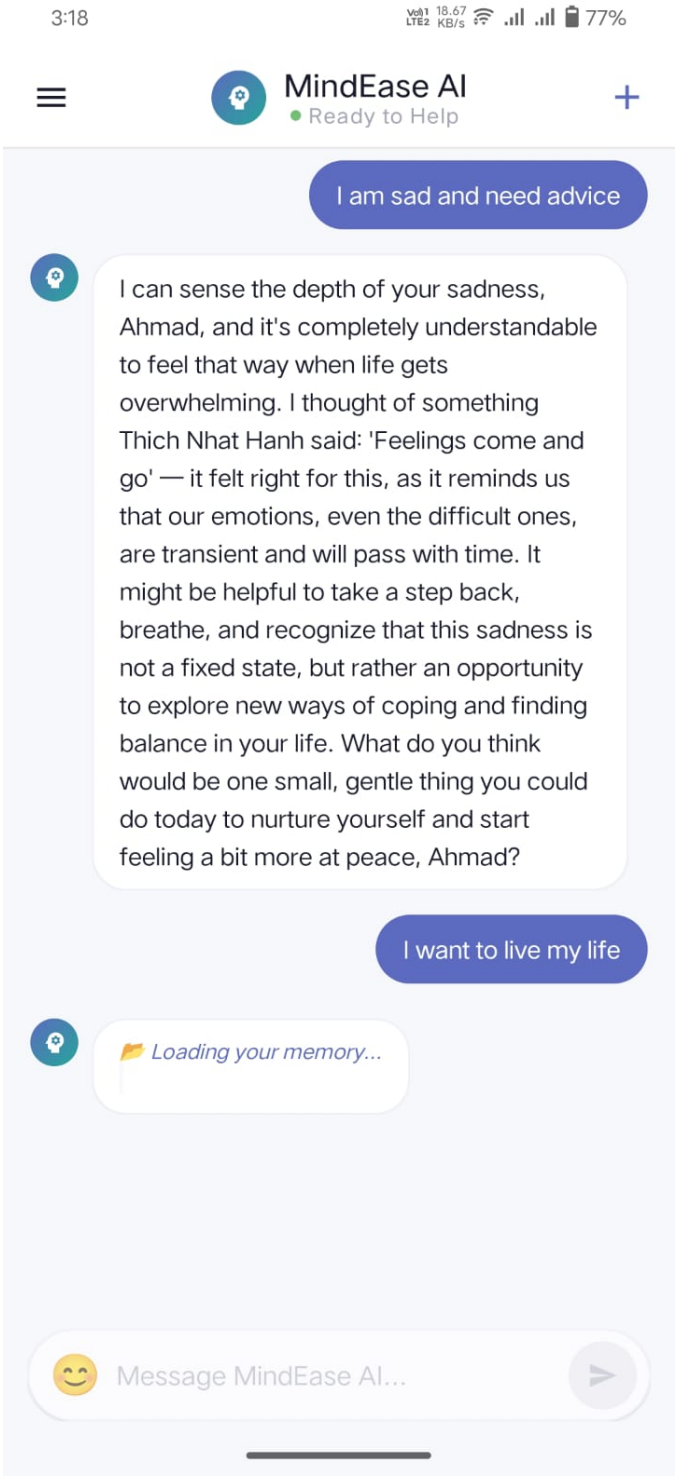


Figure 4.5: The chat screen for talking with the AI companion.

4.4.6 Mood Tracking Screen

This screen will allow users to take notes and notice their mood at various points throughout the day, and overtime can create a very basic trend line. This can enable people to see for themselves where there are patterns to be seen, instead of just relying on memory.



Figure 4.6: This screen enables people to record how they're feeling during the day.

4.4.7 Quotes Suggestion Screen

The app picks up the mood a person has been logged in with and surfaces a short quote that is supposed to reflect the user's mood. It's a small thing, but it seemed like it's a sign of paying attention to the pilot user when it shows them something relevant instead of a quote like "disconnect from the internet.Talking about the small things, pilot users noted that seeing something readily related to them, instead of some generic quote like "disconnect from the internet", felt like a gesture of respect.

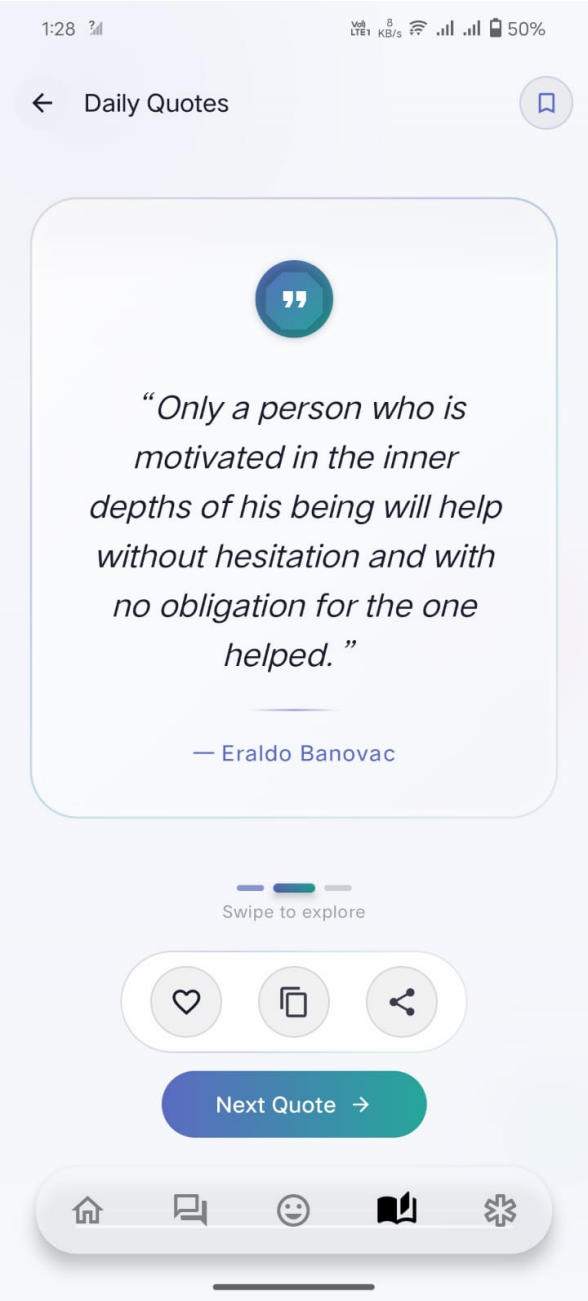


Figure 4.7: A quotation to the user depending on his/her mood.

4.4.8 Emergency Crisis Support

Here is displayed if the crisis-detection safeguard activates during a chat session, when the language emitted by the user includes a message that he/she might be in danger. It also provides helpline telephone numbers and other professional resources, as the App is not meant to be the sole means of assistance during an actual emergency.

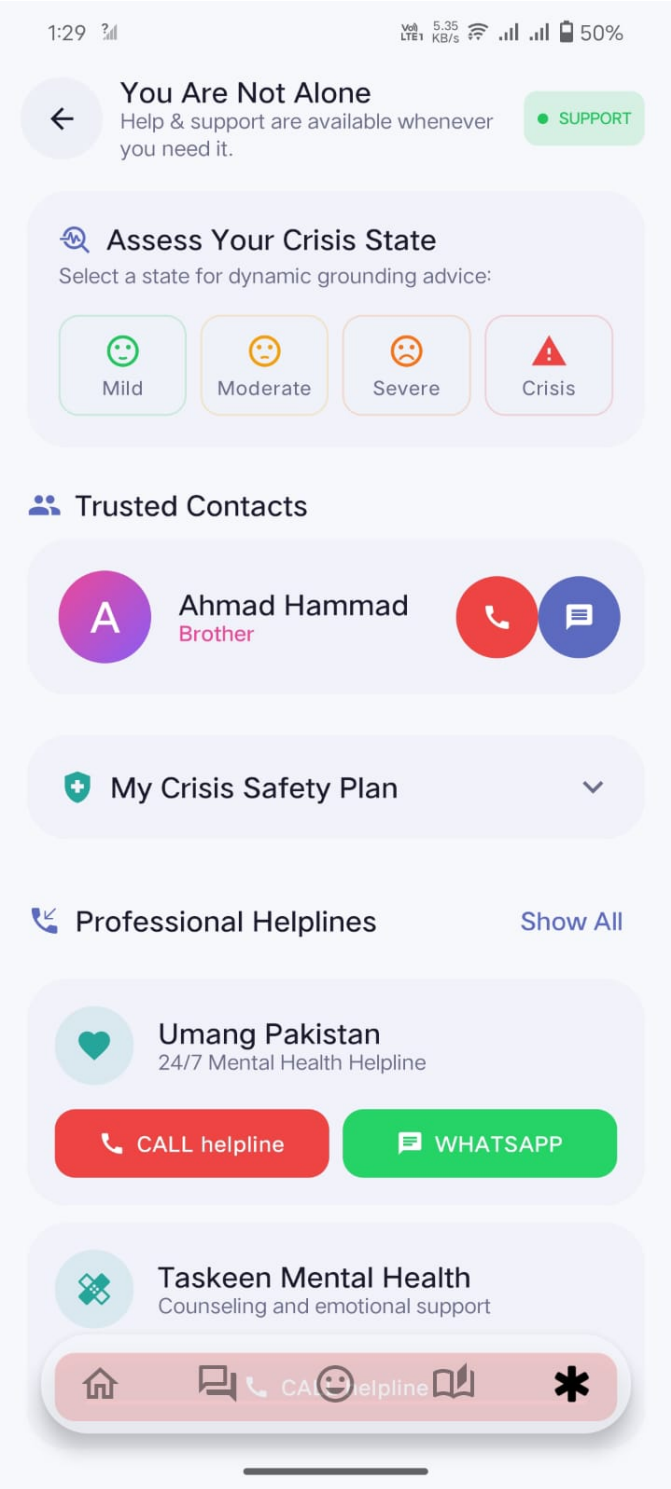


Figure 4.8: Crisis-support resources screen.

4.5 Work Plan

The results demonstrate that the implementation of the combined approach of conversational AI, emotion analysis and personalization in a single companion app is relevant and feasible. The emotion-analysis model could accurately discern the emotion used in the text written by the user, enabling the conversational engine to change its response based on the emotional tone and/or helping the personalization engine to suggest relevant topics. The users who were trying it out, reported a soothing and encouraging experience; they particularly liked the sense of privacy during the experience. The app's safety-first approach has been reinforced by the crisis-detection safeguard, since it performed well during testing as well. All these findings together give them the idea that MindEase has thecapabilities it claims to have and that, overall, the project concept and the content of the device, which is envisaged to be more capable to tailor its information and recommendations to the user, is as intended and will be more attractive to users than the generic information usually shared by apps.

4.6 Detailed Work Plan

In task distribution, the work has been split into the following team members and performed as per the timeline in Figure 3.9. After the literature review and survey, it was elaborated into a clear system design, followed by the design of the AI engine and the mobile application, with the system working concurrently and independently, and in the last phase, all the different components were integrated and tested, with user feedback gathered to validate the results.

4.7 Market Forecasting

The global digital mental-health market is expanding rapidly, with an increasing number of people now understanding the impact of mental illness and health and mental health issues becoming more routine and accepted as a form of treatment, while a chronic lack of skilled practitioners continues to drive business in the market. It is recognized that in markets such as Pakistan, there is a large group of people that is not well served, and a solution that is personalized, pays attention to privacy and affordable, can help fill in that void. A freemium system where basic support is offered at no cost and premium support is for an extra fee can be useful and is a good option to sustain growth while maintaining the essential support free for everyone who requires it. The cost for supporting an additional user in this project is relatively low and the project has a clear social purpose, so MindEase has good potential as a viable business as well as something that could benefit from continued support via grants or institutional funding.

CHAPTER 5: CONCLUSION

5.1 Conclusion

The project has been developed to address the acute under-servicing and the high prevalence of mental health problems in Pakistan, a problem that is under-reported and widely overlooked in public health efforts. To address this, MindEase provides a private, accessible, and individualized mental wellness companion that can be taken with users on their cellular phone. The motivation of the project and research on which it is based was discussed throughout this report, while user needs were explored through a market survey, a complete system design was presented that brings together four features that most existing apps offer separately; a caring, LLM-powered conversational engine, a new emotion-analysis system derived from transformer models and a personalization engine that adjusts the guidance based on the individual user's emotional state, and history, and last but not least, a new crisis-detection safeguard that links users to real human assistance should an actual crisis be detected.The World Health Organization has a test called the WHO-5 evaluation. MindEase also has a mood monitoring dashboard. This dashboard helps people see how they are doing over time. The interface is very calm and easy to use. You do not need any goggles. MindEase is a kind of support system. It uses intelligence to help people. This system is more interesting than the wellness apps. The usual apps have the old content all the time. MindEase is different because it uses intelligence. It is also affordable because it uses technology. The people who made MindEase want to help people be healthy and happy. This is related to the United Nations goals, the goal of Good Health and Well-Being which is called SDG 3. They also want to help countries develop their technology and infrastructure which is called SDG 9. MindEase is a way to provide mental health services to countries that do not have a lot of resources. The MindEase system is an example of how digital mental health services can be provided in a way that is affordable and accessible to everyone. The MindEase project shows that personalized AI-based support, like the kind used in MindEase, is not just possible, but very helpful.

5.2 Limitations

So there are some things you need to know about the system we have now. The current version of the system has a limitations that you should be aware of, with the system. You have to keep these limitations in mind when you are using the system.

- **Language support.** The app is in English at the moment. This is a problem because the people who are supposed to use the app would probably feel better if they could use it in Urdu. The app is meant for people who speak Urdu. It would be nice if the app was, in Urdu too.

- **Type of interaction.** The system is only able to understand what the user types, for now. It does not have the ability to listen to the users voice or understand what the user is feeling from their expressions. The system only supports text input.

- **MindEase is not a replacement for care.** It is meant to offer wellness guidance. MindEase is not meant to give diagnosis or treatment. People should not use MindEase as a substitute for mental health care from a doctor or therapist. MindEase is

meant to help with wellness. People should still see a doctor or therapist, for mental health care.

- **You need to be online to use this.** The AI engine requires internet access. It needs an internet connection so it can talk to the inference service. This is how the AI engine works with the inference service.
- **We have only done a testing so far.** The testing of the app was only done with a group of people so we need to do more testing with a bigger group of people, over a longer period of time to see how the app really works when people use it every day with the app.

5.3 Future Work

Looking forward MindEase is following paths to develop the product.

- MindEase is going to add support for Urdu and other languages so that more people can use it.
- MindEase will also add voice input and facial emotion recognition to make it easier for people to use MindEase. This will help people who have trouble typing or need an extra help.
- MindEase wants to help people get the support they need from a therapist. MindEase is setting up ways for people to talk to a therapist like video calls or phone consultations if they need more help than just a chat or message.
- MindEase is working on a way for the application to work on a device without an internet connection. This will also help keep peoples information safe.
- MindEase is using health experts to make sure the app works correctly. MindEase is also making sure that the app is reviewed and approved by the institutions.
- MindEase is going to introduce a community support and peer support feature. MindEase will make sure that this feature is safe and private, for everyone who uses it. MindEase will have people review and approve the community support feature to ensure it is strictly supervised.

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ABBREVIATIONS

AI : Artificial Intelligence

API : Application Programming Interface

BERT : Bidirectional Encoder Representations from Transformers

CBT : Cognitive Behavioural Therapy

CNN : Convolutional Neural Network

DFD : Data-Flow Diagram

ER : Entity-Relationship

FYDP : Final Year Design Project

FYP : Final Year Project

HEC : Higher Education Commission

JSON : JavaScript Object Notation

LLM : Large Language Model

mHealth : Mobile Health

ML : Machine Learning

NLP : Natural Language Processing

PEC : Pakistan Engineering Council

PKR : Pakistani Rupee

RAG : Retrieval-Augmented Generation

SDG : Sustainable Development Goal

UI : User Interface

UN : United Nations